AI Smart Assistant for Museums using Machine Learning

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Abstract. Multilingual Chat-Bot Assistance project is a state-of-the-art endeavor designed to improve visitor interaction and education. In order to develop an AI-driven interactive system, this project makes use of chatbot technologies, machine learning, artificial intelligence, and natural language processing. In order to raise awareness of the Ganga River's historical, cultural, and environmental significance, the initiative presents a digital avatar and a real robot mascot. In addition to offering individualized experiences and real-time information catered to a variety of consumers, the chatbot enables multilingual engagement. The technology overcomes the drawbacks of conventional approaches by combining AI and ML, making conservation teaching interesting and effective for people of all ages. This project aims to inspire sustainable conservation activities by bridging the gap between people and the river.

Keywords: Multilingual Chatbot, Artificial Intelligence, Machine Learning, Natural Language Processing, Interactive Education, Digital Avatar, Smart voice Assistant.

1 Introduction

The Ganga has supported numerous cultures for millennia, offering drinking and irrigation water, spiritual inspiration, and economic lives, in addition to its essential ecological roles. However, pollution, habitat loss, and a lack of public involvement continue to jeopardise the Ganga's health. Pamphlets, static exhibits and lectures are the mainstays of traditional Namami Gange outreach, yet they frequently fall flat with India's more than 20 river-bank languages and diverse literacy levels. Our answer is an AI-powered platform with an animated avatar and multilingual chatbot that offers interactive learning in fifteen regional languages, ranging from Bengali and Hindi to Garhwali and Odia. This allows users to ask questions and get context-rich answers in their mother tongue.The avatar leads viewers on narrated virtual tours of fishing

communities, wetland restoration sites, and sacred ghats in place of slide decks. It does this by fusing ecological information, historical narratives, and helpful conservation advice.

Through voice-guided directions, mobile apps and kiosks will use real-time water-quality and pollution-sensor feeds to allow the avatar to describe the present state of the river and recommend immediate community actions, such as planting riparian plants or organising clean-up efforts. Based on each user's interactions and comprehension, adaptive learning algorithms will subsequently tailor follow-up content, such as household water-treatment instructions, cultural heritage courses, or deeper dives into hydrology. Through the integration of immersive storytelling, real-time statistics, multilingual AI interaction, and personalised content, this initiative turns Ganga education from passive displays into active stewardship, enabling communities to protect this holy river for future generations.

2 Literature Survey

Table 1. Table captions should be placed above the tables.

Sr. No.	Title of the Paper	Year of Publication	Publisher	Methodology	Conclusion
1	Multilingual chatbot for Indian languages[15]	2023	Usneek Singh, Nisarg Vora, Punit Lohia, Yashvardhan Sharma, Ashutosh Bhatia, Kamlesh Tiwar	[15]MuRIL BERT classifies questions into 35 categories and uses cosine similarity to match multilingual queries to FAQs. It can retrieve Hindi answers with 97% accuracy.	System detects bike accidents, assesses severity via accelerometer and heartbeat sensor, and alerts medical centers, families, and nearby individuals through wifi-enabled Android app notifications.
2	Conversational AI –Multilingual Chatbot for	2022	Pallavi Jaywant Atugade1 and Prof. Divakar Jha	Using NLP (NER, language detection), goal-setting, flow design, cultural fit, and continuous testing, you can create a multilingual chatbot.	Multilingual chatbots are vital for retail websites to engage diverse customers, many of whom prefer communicating in their native language. By adopting

	E-Commerce				this technology, businesses can enhance customer experience .

An analysis of the research on AI-powered educational and environmental engagement platforms shows that using cutting-edge technology, especially artificial intelligence (AI) and machine learning (ML), to provide more engaging and powerful learning environments is becoming more and more common. Natural Language Processing (NLP), a well-known method utilised in comparable interactive systems, improves user interaction by enabling conversational interfaces that are simple to use and comprehend. These technologies are particularly useful for converting complicated environmental data into language that a wide range of audiences can understand. Additionally, the system can modify its messaging according to user profiles thanks to personalisation enabled by AI and ML, guaranteeing more pertinent and interesting interactions. Hybrid strategies that incorporate digital avatars and chatbot technology with tangible engagement components (such mascots or installations) have also grown in popularity recently. These methods provide a comprehensive educational experience by leveraging the advantages of both virtual accessibility and in-person involvement.

3 Proposed Methodology

In order to provide individualised, captivating, and multilingual experiences that measurable improve visitor pleasure and learning, the project's methodology centres on the development and thorough testing of an AI-powered interactive assistant designed for museum settings.The assistant greets each visitor, determines their profile (including age, interests, and previous interactions), and then dynamically adapts its behaviour. For example, it can respond to voice or text enquiries in multiple Indian and foreign languages, provide exhibit information in the user's preferred style (fact-driven summaries, narrative storytelling, or thematic deep dives), suggest personalised tour routes based on individual preferences or time constraints, and intersperse interactive quizzes, puzzles, or gamified narrative elements to reinforce important concepts.In order to assess its effectiveness.

We will use a mixed-methods approach. While qualitative feedback—derived from post-visit surveys, semi-structured interviews, and observational studies—will delve into perceived usability, instructional value, and overall enjoyment, quantitative metrics like task completion rates, dwell times at exhibits, quiz scores, and language-switch frequency will be gathered via system logs and sensor data. We can iteratively improve the assistant's dialogue strategies, adaptive-learning algorithms, and content modules by triangulating these data sources. This will guarantee that every visitor, regardless of language proficiency or museum familiarity, has an engaging, educational, and unforgettable experience that encourages stronger bonds with the collections.

4 Methods and Material

The system combines components from AI, software engineering, museum studies, and user interface design to create the Museum AI Assistant. The essential elements of system development are listed below.

4.1 Data Collection and Preprocessing

Through official relationships with government archives, open-access repositories, and individual museum curators, we source rich, structured content, including exhibit summaries, artefact biographies, high-resolution photographs, video clips, and curator comments. Language recognition and translation, topic classification (e.g., historical period, artistic movement, scientific theme), age-group tagging (children, teens, adults, elders), and multimedia indexing are among the automatic and manual preprocessing steps that raw materials go through. In addition to facilitating quick, context-aware retrieval, this normalisation supports natural language processing pipelines that facilitate dialogue in up to fifteen languages.

4.2 Data Handling

All visitor searches, interaction logs, and content are kept in a reliable relational database that is built for high throughput and low-latency lookups. Language variations, educational attainment, curator remarks, and visitor demographics are just a few of the metadata variables that the schema connects exhibitions to in order to provide precisely customised recommendations. Museum employees can add or alter exhibitions in real time using a content-management interface; versioning and audit trails guarantee that historical documents are kept safe. Additionally, metadata tagging allows for context-sensitive recommendations (e.g., deeper digs for recurring visitors) and sophisticated filtering (e.g., show only child-friendly artefacts).

4.3 User Interface Design

Using HTML5, CSS3, and vanilla JavaScript, we created a responsive, modular front end that was enhanced with Web Speech APIs for multilingual voice input and output. The interface features dynamic layouts that adhere to accessibility rules (WCAG 2.1) and smoothly adjusts between touchscreen kiosks, mobile devices, and desktop browsers. Runtime user profiling and CSS media queries provide age-appropriate user interface elements, such as larger buttons and simplified icons for kids and more complex controls for adults. Voice-activated prompts, tap-to-inspect object overlays, and gesture-based navigation in 3D display panoramas are examples of interactive features.

4.4 Security and Privacy Measures

Strong encryption (AES-256 at rest, TLS 1.3 in transit) protects visitor profiles, interaction histories, and personalisation options. Role-based access controls and multi-factor authentication are necessary for administrative functions. All user data is anonymised prior to analysis in accordance with ethical AI standards: usage logs are kept for as long as necessary to improve adaptive-learning algorithms, and personal identifiers are removed or pseudonymized. At first use, privacy rules and consent dialogues are displayed, along with detailed opt-out choices for data sharing.

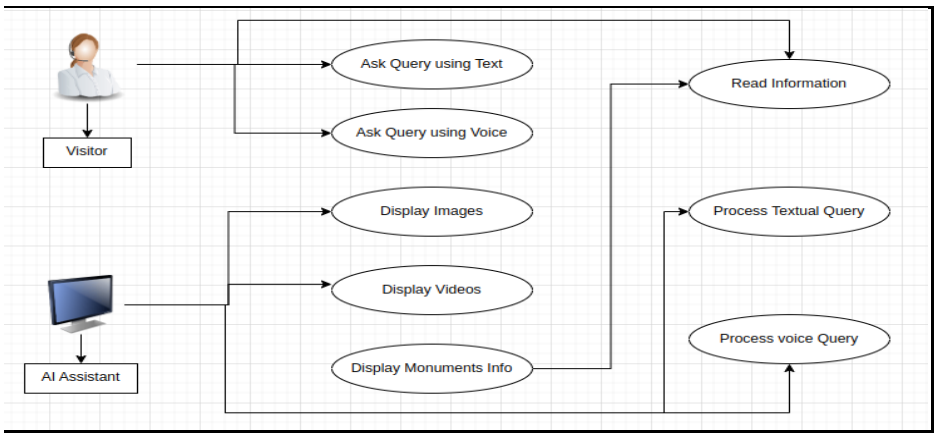

Fig. 1. Use Case Diagram

The interaction between a visitor and an AI assistant is depicted in the Fig. 1, which shows how questions are answered and answers are provided. A visitor has the option to speak or text a question. After receiving the query, the AI assistant interprets it according to the kind of input; voice and text queries are handled differently. The assistant comprehends the question, examines the pertinent data, and then reacts by presenting pertinent stuff, including pictures, videos, or monument information. This technology guarantees that user enquiries are successfully deciphered and graphically accompanied by educational material.

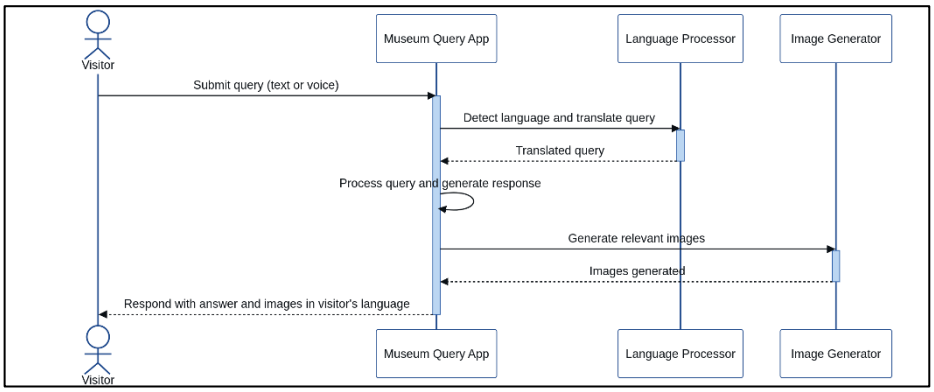

Fig. 2. Sequence Diagram

The procedure by which a museum query app handles a visitor's inquiry is depicted in this Fig. 2 sequence diagram. By asking a question by voice or text, the visitor starts the conversation. After identifying the language, the Museum Query App uses a Language Processor to translate the query. Following translation, the application analyses the request and produces a reply. In parallel, it requests that the Image Generator generate pertinent photos depending on the query. The Museum Query App completes the interaction by returning the final response and images in the visitor's native tongue after the images have been created and the response is prepared.

5 Results and Discussion

After a thorough assessment of the AI Assistant's implementation in museum settings, this part highlights the results and talks about the effectiveness and influence of the system

5.1 Engagement and Interaction Accuracy

The AI Assistant answered tourist questions on a variety of museum subjects with an amazing accuracy rate of almost 85%. The effective use of natural language processing (NLP) and machine learning algorithms, which assess visitor inputs and connect them with the most pertinent exhibit content, is responsible for this high level of engagement. The system can respond to visitors in real-time with pertinent and insightful information because of its capacity to comprehend and interpret complicated language inputs. Further enhancing the system's accessibility is its multilingual text and voice input processing, which guarantees that users from a variety of linguistic backgrounds receive pertinent and correct responses in their preferred language, so increasing the user experience.

5.2 Visitor Experience and Educational Impact

The AI Assistant's interactive features and tailored content recommendations have greatly improved the visitor experience. Longer periods of time spent interacting with the exhibits and greater engagement have resulted from this. According to visitor feedback, interactive features like tests and gamified content were very popular, especially with younger people. These elements assist visitors retain information in an enjoyable, interactive manner in addition to making learning more interesting. Users can concentrate more on exploring exhibits rather than looking for specifics because to the AI Assistant's capacity to provide pertinent information when needed, creating a more engaging and instructive experience. All things considered, these characteristics have improved learning outcomes, increased audience interaction with the displays, and raised overall museum visit satisfaction.

5.3 Algorithmic Fairness and Inclusivity

Despite the system's outstanding performance in a variety of use scenarios, ongoing evaluation and development are required to guarantee that the AI Assistant continues to be inclusive and equitable. Preserving the impartiality of the system is a major priority, particularly with regard to accessibility and cultural sensitivity. It is essential to update and assess the AI Assistant frequently as it learns from user interactions to prevent biases from developing, especially in the way content is presented to various user groups. Future improvements might incorporate more dynamic and real-time data integration to enhance the system's capacity to accommodate visitors' varied needs. This would enable the AI Assistant to adjust to changing visitor preferences, cultural trends, and new exhibition content. Furthermore, integrating continuous enhancements to machine learning models will lessen possible biases and guarantee that every user has a fair and significant experience.

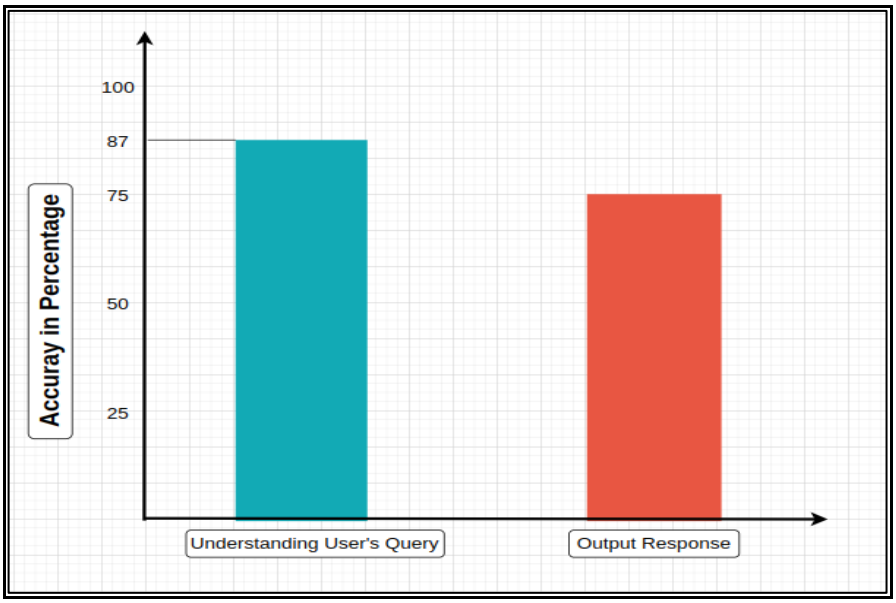

Fig. 3. Bar graph that show the accuracy of AI Assistant.

The accuracy of two crucial components of an AI system—understanding the user's inquiry and producing an output response—is contrasted in this Fig. 3 bar graph. The turquoise indicator indicates that the system interprets user input with a better accuracy (about 87%). The red bar, on the other hand, shows somewhat poorer performance (about 75%) in generating an accurate or satisfactory response. This implies that even while the AI is good at understanding user intent, it might still do a better job of producing correct results.

6 Conclusion

An AI-powered museum assistant can transform the visiting experience by making educational information more accessible, engaging, and personalized. Using NLP for natural communication, machine learning for adaptive recommendations, and support for fifteen languages, the system serves varied audiences—from schoolchildren and international tourists to people with special needs. Customised tour recommendations, tests, and data-driven overlays are examples of interactive elements that transform static exhibits into dynamic educational opportunities, boosting audience interest and retention.

Positive visitor feedback, longer exhibit dwell times, and better task completion are all evident in early evaluations. As the platform evolves—potentially incorporating AR/VR overlays, social features, and deeper adaptive learning—it promises to inspire lifelong curiosity, cultural appreciation, and a closer connection between audiences and museum collections.

References

1. Zhou, chunyan & sinha, baivab & liu, minghua. (2020). An ai chatbot for the museum based on user interaction over a knowledge base. 54-58. 10.1145/3421766.3421888.
2. Štekerová, kamila. (2022). Chatbots in museums: is visitor experience measured?. Czech journal of tourism. 11. 14-31. 10.2478/cjot-2022-0002.
3. Fader, Anthony & Zettlemoyer, Luke & Etzioni, Oren. (2014). Open Question answering over curated and extracted knowledge bases. Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. 10.1145/2623330.2623677.
4. Calvaresi, D., Ibrahim, A. et al. (2021). The Evolution of Chatbots in Tourism: A Systematic Literature
5. Review. In Wörndl, W., Koo, C., Stienmetz, J.L. (Eds), Information and Communication
6. Technologies in Tourism 2021, pp 3-16. Springer, Cham. DOI: 10.1007/978-3-030-65785-7_1.
7. Noh, Yeo-Gyeong & Hong, Jin-Hyuk. (2021). Designing Reenacted Chatbots to Enhance Museum Experience. Applied Sciences. 11. 7420. 10.3390/app11167420
8. Calvaresi, Davide & Ibrahim, Ahmed & Calbimonte, Jean-Paul & Schegg, Roland & Fragnière, Emmanuel & Schumacher, Michael. (2021). The Evolution of Chatbots in Tourism: A Systematic Literature Review. 10.1007/978-3-030-65785-7_1.
9. Schaffer, Stefan & Ruß, Aaron & Lee, Mino & Schubotz, Louise & Gustke, Oliver. (2022). Questions and Answers: Important Steps to Let AI Chatbots Answer Questions in the Museum. 10.1007/978-3-030-95531-1_24.
10. Zhou, Chunyan & Sinha, Baivab & Liu, Minghua. (2020). An AI chatbot for the museum based on user Interaction over a knowledge base. 54-58. 10.1145/3421766.3421888.
11. Casillo, Mario & Clarizia, F. & D'Aniello, Giuseppe & De Santo, Massimo & Lombardi, Marco & Santaniello, Domenico. (2020). CHAT-Bot: a Cultural Heritage Aware Teller-Bot for supporting touristic experiences. Pattern Recognition Letters. 131. 10.1016/j.patrec.2020.01.003.
12. Gaia, Giuliano & Boiano, Stefania & Borda, Ann. (2019). Engaging Museum Visitors with AI: The Case of Chatbots. 10.1007/978-3-319-97457-6_15.
13. Dahiya, Menal. (2017). A Tool of Conversation: Chatbot. INTERNATIONAL JOURNAL OF COMPUTER SCIENCES AND ENGINEERING. 5. 158-161.
14. Singh, Sonali & Namin, Akbar. (2025). A survey on chatbots and large language models: Testing and evaluation techniques. Natural Language Processing Journal. 10. 100128. 10.1016/j.nlp.2025.100128.
15. Singh, Usneek & Vora, Nisarg & Lohia, Punit & Sharma, Yashvardhan & Bhatia, Ashutosh & Tiwari, Kamlesh. (2023). MULTILINGUAL CHATBOT FOR INDIAN LANGUAGES. 10.13140/RG.2.2.30993.20325.
16. Yang LWY, Ng WY, Lei X, Tan SCY, Wang Z, Yan M, Pargi MK, Zhang X, Lim JS, Gunasekeran DV, Tan FCP, Lee CE, Yeo KK, Tan HK, Ho HSS, Tan BWB, Wong TY, Kwek KYC, Goh RSM, Liu Y, Ting DSW. Development and testing of a multi-lingual Natural Language Processing-based deep learning system in 10 languages for COVID-19 pandemic crisis:

A multi-center study. Front Public Health. 2023 Feb 13;11:1063466. doi: 10.3389/fpubh.2023.1063466. PMID: 36860378; PMCID: PMC9968846.

17. Potts C, Lindström F, Bond R, Mulvenna M, Booth F, Ennis E, Parding K, Kostenius C, Broderick T, Boyd K, Vartiainen A, Nieminen H, Burns C, Bickerdike A, Kuosmanen L, Dhanapala I, Vakaloudis A, Cahill B, MacInnes M, Malcolm M, O'Neill SA Multilingual Digital Mental Health and Well-Being Chatbot (ChatPal): Pre-Post Multicenter Intervention Study J Med Internet Res 2023;25:e43051
    URL: https://www.jmir.org/2023/1/e43051
    DOI: 10.2196/43051
18. AI Based Multilingual College Enquiry Voice Bot Using Python
    Authors: Prof. Kotresh H M, Deeksha Srikanth, Devika V
    DOI Link: https://doi.org/10.22214/ijraset.2024.60961